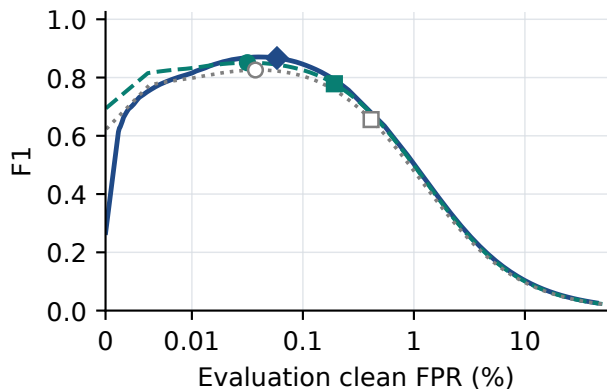


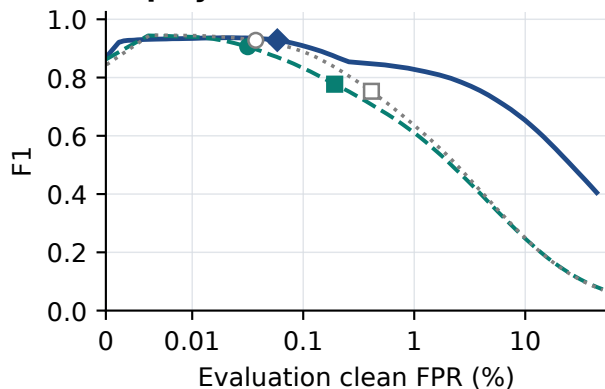
Modena: operating-point sensitivity

Calibration selects thresholds. FPR: false-positive rate; LGB: LightGBM.

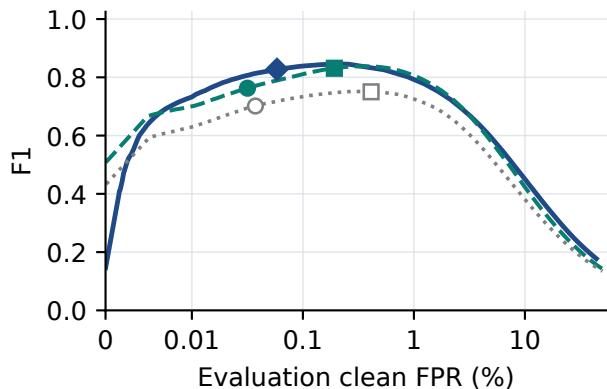
Pooled F1



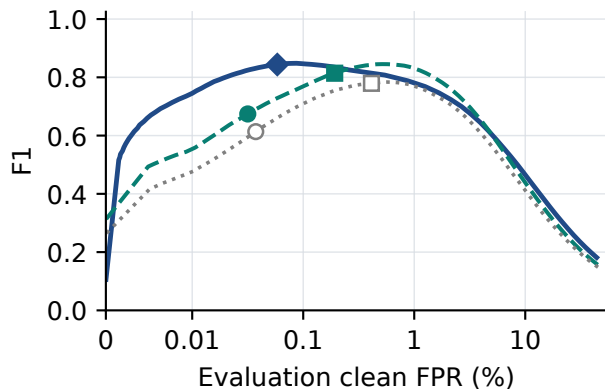
Replay F1



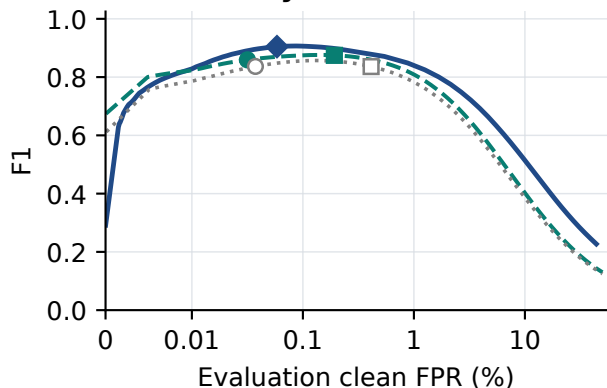
Drift F1



Noise F1



Mean family F1



- Hybrid: fixed gates/allocation
- LightGBM: 3-hour allowance
- LightGBM: zero hours
- Original hybrid
- Original LGB3h: pooled
- Original LGB3h: balanced
- Original LGB0h: pooled
- Original LGB0h: balanced

Means: 10 models $\times$ 6 shared sources. The 60 cells are not independent.

Hybrid gates/allocation fixed. Three-hour control primary; zero-hour contextual.

FPR axis: linear to 0.01%, logarithmic above. No exhaustive frontier.